Utilizing AI Threat Resources

CompTIA SecAI+ CertMaster Study — Lesson 2.1.3

Why AI Threat Resources Matter

As AI usage increases across different use cases within an organization, it increases the organization's attack surface and attack vectors. Security teams must analyze the risks not only related to the technical aspects but also behavioral, data-driven, and ethical. There is a need for referencing AI threats over time and the defense strategies used for mitigating them.

AI Threat Repositories Overview

To address the challenges of identifying threats, several AI threat repositories have been developed. These repositories consist of known AI vulnerabilities, real-time incidents, and research findings. They help security teams identify and map threats in the model development lifecycle, such as data collection, training models, and user interaction.

Key Resources Include

- MIT AI Risk Repository
- Common Vulnerabilities and Exposures (CVE) system
- AI Vulnerability Database (AVID)
- AI Incident Database (AIID)
- arXiv
- Common Weakness Enumeration (CWE)

MIT AI Risk Repository

1,600+

AI Risks

Sourced from existing frameworks

65

Frameworks

Contributing source frameworks

7

Domains

Risk classification domains

24

Subdomains

Detailed risk subdomains

The MIT AI Risk Repository is one of the major sources of AI threats. It classifies AI risks into taxonomies based on how, when, why, and in what domain the risks occurred. The repository is updated regularly and acts as a live database.

MIT Repository: Three Key Parts

1

AI Risk Database

A centralized database with over **2,244 records**, each defining an AI risk. It includes titles, keywords, evidence, and risk categories with detailed explanations.

2

Causal Taxonomy of AI Risks

Classifies risks based on how they originated, whether caused by AI, human users/developers, or external factors. It also categorizes risks by timing during the model development lifecycle.

3

Domain Taxonomy of AI Risks

Classifies risks into **seven AI risk domains** and 23 subdomains, including Discrimination & Toxicity, Privacy & Security, Misinformation, Malicious Actors & Misuse, Human-Computer Interaction, Socioeconomic & Environmental, and AI System Safety, Failures & Limitations.

CVE AI Workgroup

The CVE AI Workgroup is part of the **Common Vulnerability Enumeration (CVE) program**, composed of CVE board members, corporate members in vulnerability management, AI community members, and their associations. This working group develops policies and guidelines for identifying and reporting AI-based vulnerabilities. They analyze and identify AI threats, assigning them a **CVE ID** and making them part of the CVE database.

CVE Board Members

Core governance and oversight

Corporate Members

Vulnerability management expertise

AI Community Members

Domain-specific AI knowledge

CVE AI Workgroup: Taxonomies and Standards

The group is creating clear criteria, taxonomies, and best practices for documenting AI risks so they can be shared across AI and cybersecurity communities. This includes integrating AI-specific entries into the broader CVE framework.

Cause-Based Taxonomies

- Data poisoning
- Algorithmic errors

Domain-Based Taxonomies

- Computer vision
- Natural language processing

This allows researchers, developers, and security analysts to understand the nature of each risk and its impacts.

AI Vulnerability Database (AVID)

The **AI Vulnerability Database (AVID)** is an open-source knowledge base that collects data about failure modes for AI models, datasets, and systems. It provides a structured approach for security teams to identify, assess, and mitigate AI-specific vulnerabilities.

Identify

Search AVID for known vulnerabilities that may affect the model, dataset, or API before deployment.

Assess

Evaluate the severity and relevance of discovered vulnerabilities to your specific AI components.

Mitigate

Apply structured remediation strategies for AI-specific vulnerabilities before deployment.

AI Incident Database (AIID)

The **AI Incident Database** is a collection of real-time AI incidents from the Internet, covering everyday threats caused by AI systems like deepfakes, biases, and misuse of AI.

How It Helps Security Teams

- Study incidents and identify common attack patterns
- Understand vulnerabilities in their own systems
- Analyze how attackers target AI models
- Assess AI systems for weaknesses in data handling, training processes, or model deployment

arXiv

arXiv is an open-source platform where researchers share scientific papers covering fields including artificial intelligence. Researchers worldwide upload their latest findings to share knowledge and get feedback from the community.

AI threat resources provide insights into how threats have been identified in the past, what weaknesses may exist, and how to prevent or remediate identified issues. By using this information, developers and security teams can improve **safe, reliable, and responsible AI systems**.

What arXiv Provides

- Latest AI research findings
- Global scientific community contributions
- Peer feedback and knowledge sharing
- Insights into past threat identification
- Guidance on prevention and remediation

Common Weakness Enumeration (CWE)

The **Common Weakness Enumeration (CWE)** is a widely used framework for identifying and categorizing software vulnerabilities. For AI security, CWE maps real-world threats to specific types of system weaknesses.

CWE-77: Command Injection

Relevant for **prompt injection attacks** targeting AI systems and language models.

CWE-200: Exposure of Sensitive Information

Aligns with risks like **system prompt leakage** or model inversion attacks.

Using CWE helps **standardize the threat landscape** for AI systems and supports integration with frameworks like MITRE ATT&CK or CVE databases.

Key Takeaways

- AI threat repositories provide **known vulnerabilities, real-time incidents, and research findings** to help security teams identify and map threats across the AI development lifecycle.
- The **MIT AI Risk Repository** contains over 1,600 risks across seven domains with causal and domain taxonomies for structured risk classification.
- The **CVE AI Workgroup** develops policies and guidelines for identifying, reporting, and categorizing AI-based vulnerabilities within the CVE framework.
- **AVID** offers an open-source knowledge base for AI model, dataset, and system failure modes.
- The **AI Incident Database** catalogs real-world AI incidents to help identify attack patterns and system weaknesses.
- **arXiv** provides the latest AI research findings from the global scientific community.
- **CWE** maps real-world AI threats to specific system weaknesses, standardizing the AI threat landscape.